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Education & Credentials

- MEng: Electrical & Computer Engineering, UOWM Graduated 2024
- Certificate of Proficiency (C2) in English 2024
- Driving License 2019

Other personal info

- Military obligations Fulfilled 02/2026
- GitHub: github.com/kgewrg
- LinkedIn: linkedin.com/in/konstantinos-georgitsaros
- Portfolio: kgewrg.github.io/portfolio

Projects

Participation on a robotics University team, Hyperion Robotics

- My primary part on the team was to design and implement computer vision on the robots and allow them to recognize objects, find their position on the playing area and recognize enemy robots.
- Used cheap web cameras and computer vision (openCV) and NumPy to process the image data for Python to develop algorithms that allowed the robots to identify objects.
- Created and maintained the GitHub repository and setup git for the projects, allowing the team to easily share the codebase, also helped the team with my Linux knowledge and established a workflow making it easier for the team to work remotely on the Raspberry Pi boards driving the robots.
- Collaborated with other teams members to design and program functions, that were compatible with the mechanical constraints and capabilities of the robot's design, allowing the robots to score more points.
- Implemented ROS on the robots so each discrete part can communicate with each other.
- Used a soldering iron and wiring diagrams to correctly wire the robot's components.

KTEL planner

- An app for Android that finds itineraries with minimal wait time for the travelers that use the bus station KTEL Macedonia as a middle stop.
- Used web scrapping Python libraries to retrieve the itinerary details and designed an algorithm that combines them to find the travel plan with minimal wait time.
- Translated the algorithm to Dart and used the Flutter framework to build an Android app and published it on GitHub.

- Used Express.js and Node.js framework to build an REST API for the algorithm and with HTML, CSS and JavaScript made a user interface.
- Used GitHub actions and hosting websites to make the app available to public.

Diploma Thesis “Brain de fer”

- Videogame that allows two players to compete against each other based on how focused they are.
- Used electroencephalograph brain-computer interfaces to retrieve brain activity signals and developed Python algorithms to derive the focus “amount” from those signals.
- Used the Unity game engine to design and build a user interface for the players.
- Build an interface between Python and C# to pass the data from the electroencephalographs to the game engine.
- Presented the thesis as a university exhibition at 87th TIF, and with my remote support the university presented the thesis on other events and fairs.

Air hockey, semester school project

- A small air hockey table that a player can play against an automatic system
- Used cheap web cameras and computer vision to find the location of the puck and implemented in Python a algorithm to predict the pucks path so the striker can intercept it.
- Implemented a serial interface in Python so the algorithm can communicate with the Arduino Mega controlling the stepper motors.
- Designed, 3d printed and machined structural parts for the kinetic system, providing a stiff and robust base for the stepper motors.
- Used a soldering iron to wire the Arduino Mega microcontroller boards to the stepper drivers and motors and tuned the kinematic system resulting a fast and responsive movement of the striker.

Work History

Smartphone repair technician, PowerPC, Kilkis

02/2024 – 02/2025

- *Managed repair priority and carried them out. Communicated with the customer and parts providers, always trying to provide a top quality and quick repair.*
- Repaired smartphones by troubleshooting software and hardware problems, mainly by changing phone parts but often by doing board-level repair, giving lots of attention to detail to ensure a quality repair.
- Handled high-volume periods trying to execute urgent repairs first, worked on multiple repairs simultaneously, utilized ERP software in order to guarantee good customer service.
- Managed the parts inventory by calling and synergizing with providers making sure we always had good quality parts for most common repairs.

- Repaired and maintained tank simulators through microcontroller programming and precision soldering, while simultaneously maintaining network and domain infrastructure.
- Integrated and programmed microcontrollers for tank simulators by wiring and soldering control handles, resulting in improved system responsiveness and simulation accuracy.
- Diagnosed and repaired low-level simulator hardware and peripheral components to ensure 100% operational readiness for training exercises.
- Managed Windows Domain environments by configuring user accounts and security groups to maintain secure and organized access control.
- Troubleshoot and maintained local area network (LAN) infrastructure and hardware to minimize downtime for administrative operations.
- Optimized computer hardware performance through regular maintenance and assisted staff with IT support such as data recovery on external drives and repairing multifunction printers (MFPs).